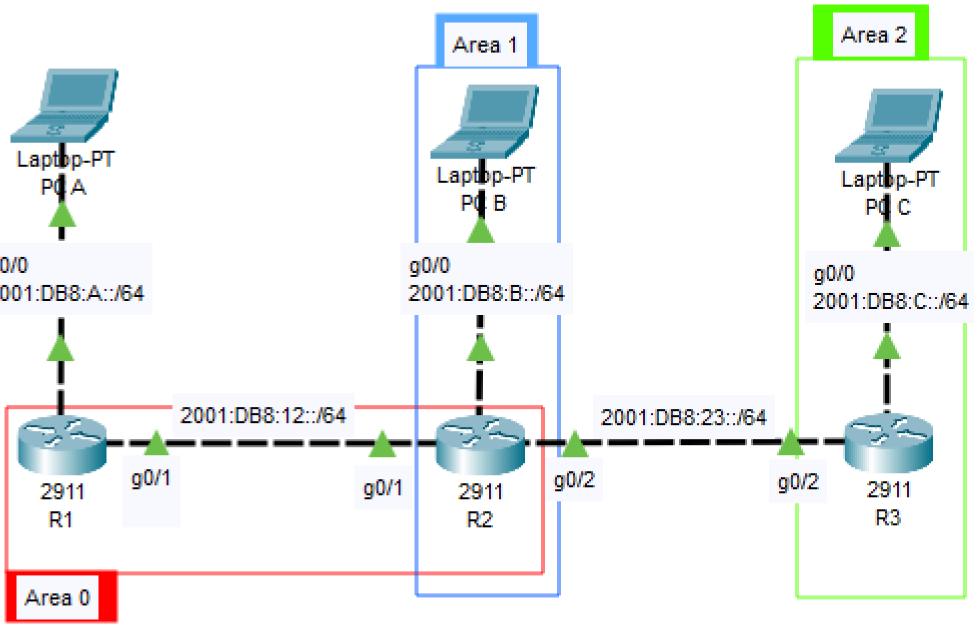
Configuration d'OSPFv3 Multi-Area pour IPv6

Sommaire

Schéma de la configuration à réaliser



[PC-A] – (G0/0: 2001:DB8:A::/64) -- [Router1]

(Area 0)
 [Router1] -- (G0/1: 2001:DB8:12::/64) -- [Router2]
 [Router2] -- (G0/1: 2001:DB8:23::/64) -- [Router3]

(Area 1)
 [PC-B] – (G0/0: 2001:DB8:B::/64) -- [Router2]

(Area 2)
 [PC-C] – (G0/0: 2001:DB8:C::/64) -- [Router3]

(Area 2)
Aires (Areas) :
Area 0 (Backbone) : Lien entre Router1 et Router2.
Area 1 : Lien entre Router2 et PC-B.
Area 2 : Lien entre Router3 et PC-C.
Adresses IPv6 : Préfixes /64 pour tous les liens.

Étape 1 : Activer IPv6 et configurer les interfaces

Sur tous les routeurs, activez le routage IPv6 et configurez les adresses.

```
R~(config)# ipv6 unicast-routing
```

Router1 :

```
G0/0 : 2001:DB8:A::1/64 (vers PC-A)
G0/1 : 2001:DB8:12::1/64 (vers Router2, Area 0)
```

```
R1#show ipv6 interface brief
GigabitEthernet0/0         [up/up]
    FE80::203:E4FF:FECB:501
    2001:DB8:A::1
GigabitEthernet0/1         [up/up]
    FE80::203:E4FF:FECB:502
    2001:DB8:12::1
GigabitEthernet0/2         [up/down]
    unassigned
Vlan1                      [administratively down/down]
    unassigned
R1#
```

Router2 :

```
G0/1 : 2001:DB8:12::2/64 (Area 0)
G0/0 : 2001:DB8:B::1/64 (vers PC-B, Area 1)
G0/2 : 2001:DB8:23::2/64 (vers Router3, Area 0)
```

```
R2#Show ipv6 interface brief
GigabitEthernet0/0          [up/up]
    FE80::250:FFF:FE51:E101
    2001:DB8:B::1
GigabitEthernet0/1          [up/up]
    FE80::250:FFF:FE51:E102
    2001:DB8:12::2
GigabitEthernet0/2          [up/up]
    FE80::250:FFF:FE51:E103
    2001:DB8:23::2
Vlan1                       [administratively down/down]
    unassigned
R2#
```

Router3 :

```
G0/2 : 2001:DB8:23::3/64
G0/0 : 2001:DB8:C::1/64 (vers PC-C, Area 2)
```

```
R3#show ipv6 interface brief
GigabitEthernet0/0          [up/up]
    FE80::201:42FF:FE50:C01
    2001:DB8:C::1
GigabitEthernet0/1          [up/down]
    unassigned
GigabitEthernet0/2          [up/up]
    FE80::201:42FF:FE50:C03
    2001:DB8:23::3
Vlan1                       [administratively down/down]
    unassigned
R3#
```

Étape 2 : Vérifier la connectivité locale

Depuis Router1, pinguez Router2.

```
R1#ping 2001:DB8:12::2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:12::2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1#
```

Étape 3 : Activer OSPFv3 sur les routeurs

OSPFv3 est configuré par interface (contrairement à OSPFv2 qui utilise des network statements).

Sur Router1 (Area 0) :

```
Router1(config)# ipv6 router ospf 1
Router1(config-rtr)# router-id 1.1.1.1 (ID unique format IPv4)
Router1(config-rtr)# exit
```

```
Router1(config)# interface GigabitEthernet0/0
Router1(config-if)# ipv6 ospf 1 area 0 (PC-A dans Area 0 Backbone)
Router1(config-if)# interface GigabitEthernet0/1
Router1(config-if)# ipv6 ospf 1 area 0  (Lien vers Router2 Area 0)
```

```
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
 ipv6 address 2001:DB8:A::1/64
 ipv6 ospf 1 area 0
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 ipv6 address 2001:DB8:12::1/64
 ipv6 ospf 1 area 0
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
```

Sur Router2 (Area 1) :

```
Router2(config)# ipv6 router ospf 1
Router2(config-rtr)# router-id 2.2.2.2 (ID unique format IPv4)
Router2(config-rtr)# exit
Router2(config)# interface GigabitEthernet0/0
Router2(config-if)# ipv6 ospf 1 area 1 (PC-B dans Area 1)
Router2(config)# interface GigabitEthernet0/1
Router2(config-if)# ipv6 ospf 1 area 0 (Lien vers Router1 Area 0)
Router2(config-if)# interface GigabitEthernet0/2
Router2(config-if)# ipv6 ospf 1 area 0  (Lien vers Router3 Area 0)
```

```
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
 ipv6 address 2001:DB8:B::1/64
 ipv6 ospf 1 area 1
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 ipv6 address 2001:DB8:12::2/64
 ipv6 ospf 1 area 0
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 ipv6 address 2001:DB8:23::2/64
 ipv6 ospf 1 area 2
```

Sur Router3 (Area 2) :

```
Router3(config)# ipv6 router ospf 1
Router3(config-rtr)# router-id 3.3.3.3 (ID unique format IPv4)
```

```
Router3(config-rtr)# exit
Router3(config)# interface GigabitEthernet0/0
Router3(config-if)# ipv6 ospf 1 area 2 (PC-C dans Area 2)
Router3(config-if)# interface GigabitEthernet0/2
Router3(config-if)# ipv6 ospf 1 area 2  (Lien vers Router2 Area 2)
```

```
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
 ipv6 address 2001:DB8:C::1/64
 ipv6 ospf 1 area 2
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 ipv6 address 2001:DB8:23::3/64
 ipv6 ospf 1 area 2
!
```

Étape 4 : Vérifier les voisins OSPFv3

```
R1# show ipv6 ospf neighbor
```

```
R1#show ipv6 ospf neighbor

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
2.2.2.2           1   FULL/BDR        00:00:38    2               GigabitEthernet0/1
R1#
```

Sur Router2, affichez les voisins OSPFv3.

```
R2# show ipv6 ospf neighbor
```

```
R2#show ipv6 ospf neighbor

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
1.1.1.1           1   FULL/DR         00:00:31    2               GigabitEthernet0/1
3.3.3.3           1   FULL/BDR        00:00:33    3               GigabitEthernet0/2
R2#
```

```
R3# show ipv6 ospf neighbor
```

```
R3#show ipv6 ospf neighbor

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
2.2.2.2           1   FULL/DR         00:00:31    3               GigabitEthernet0/2
R3#
```

Étape 5 : Afficher la base de données LSDB

Sur Router1 :

```
Router1# show ipv6 ospf database
```

```
R1#show ipv6 ospf database
            OSPF Router with ID (1.1.1.1) (Process ID 1)

                Router Link States (Area 0)

ADV Router      Age         Seq#        Fragment ID  Link count Bits
1.1.1.1         240         0x80000002 0             1
2.2.2.2         215         0x80000003 0             2
3.3.3.3         215         0x80000002 0             1

                Net Link States (Area 0)
ADV Router      Age         Seq#        Link ID (DR)    Rtr count
1.1.1.1         240         0x80000001 2                2
2.2.2.2         215         0x80000001 3                2

                Link (Type-8) Link States (Area 0)
ADV Router      Age         Seq#        Link ID     Interface
1.1.1.1         1225        0x80000001 1            Gi0/0
1.1.1.1         244         0x80000003 2            Gi0/1
2.2.2.2         240         0x80000004 2            Gi0/1

                Intra Area Prefix Link States (Area 0)
ADV Router      Age         Seq#        Link ID     Ref-lstype  Ref-LSID
1.1.1.1         240         0x80000003 1            0x2002      2
1.1.1.1         240         0x80000004 2            0x2001      0
3.3.3.3         217         0x80000002 2            0x2001      0
2.2.2.2         215         0x80000004 1            0x2002      3
2.2.2.2         215         0x80000005 2            0x2001      0
R1#
```

Sur Router2 :

```
Router2# show ipv6 ospf database
```

```
R2#show ipv6 ospf database
            OSPF Router with ID (2.2.2.2) (Process ID 1)

                Router Link States (Area 0)

ADV Router      Age         Seq#        Fragment ID  Link count Bits
1.1.1.1         169         0x80000002 0             1
2.2.2.2         144         0x80000003 0             2
3.3.3.3         144         0x80000002 0             1

                Net Link States (Area 0)
ADV Router      Age         Seq#        Link ID (DR)    Rtr count
1.1.1.1         169         0x80000001 2                2
2.2.2.2         144         0x80000001 3                2

                Link (Type-8) Link States (Area 0)
ADV Router      Age         Seq#        Link ID     Interface
2.2.2.2         419         0x80000001 1            Gi0/0
2.2.2.2         169         0x80000004 2            Gi0/1
1.1.1.1         173         0x80000003 2            Gi0/1
2.2.2.2         146         0x80000005 3            Gi0/2
3.3.3.3         144         0x80000003 3            Gi0/2

                Intra Area Prefix Link States (Area 0)
ADV Router      Age         Seq#        Link ID     Ref-lstype  Ref-LSID
1.1.1.1         169         0x80000003 1            0x2002      2
1.1.1.1         169         0x80000004 2            0x2001      0
3.3.3.3         146         0x80000002 2            0x2001      0
2.2.2.2         144         0x80000004 1            0x2002      3
2.2.2.2         144         0x80000005 2            0x2001      0
R2#Z
```

Sur Router3 :

```
Router3# show ipv6 ospf database
```

```
R3#show ipv6 ospf database
                OSPF Router with ID (3.3.3.3) (Process ID 1)

                     Router Link States (Area 0)

ADV Router       Age          Seq#        Fragment ID  Link count Bits
1.1.1.1          264          0x80000002 0             1
3.3.3.3          239          0x80000002 0             1
2.2.2.2          239          0x80000003 0             2

                     Net Link States (Area 0)
ADV Router       Age          Seq#        Link ID (DR)    Rtr count
1.1.1.1          264          0x80000001 2                2
2.2.2.2          239          0x80000001 3                2

                     Link (Type-8) Link States (Area 0)
ADV Router       Age          Seq#        Link ID     Interface
3.3.3.3          459          0x80000001 1           Gi0/0
3.3.3.3          239          0x80000003 3           Gi0/2
2.2.2.2          241          0x80000005 3           Gi0/2

                     Intra Area Prefix Link States (Area 0)
ADV Router       Age          Seq#        Link ID     Ref-lstype  Ref-LSID
3.3.3.3          241          0x80000002 2           0x2001      0
1.1.1.1          264          0x80000003 1           0x2002      2
1.1.1.1          264          0x80000004 2           0x2001      0
2.2.2.2          239          0x80000004 1           0x2002      3
2.2.2.2          239          0x80000005 2           0x2001      0
R3#
```

Etape 6 : Tests de connectivité inter-areas

Configuration de L'IP de PC-A

IPv6 Configuration

○ Automatic	◉ Static
IPv6 Address	2001:DB8:A::10 / 64
Link Local Address	FE80::230:F2FF:FE6B:B429
Default Gateway	2001:DB8:A::1
DNS Server	

Configuration de L'IP de PC-C

IPv6 Configuration

○ Automatic	◉ Static
IPv6 Address	2001:DB8:C::20 / 64
Link Local Address	FE80::240:BFF:FE33:4774
Default Gateway	2001:DB8:C::1
DNS Server	

Depuis PC-A, pinguez PC-C .

```
C:\>ping 2001:DB8:C::20

Pinging 2001:DB8:C::20 with 32 bytes of data:

Reply from 2001:DB8:C::20: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:C::20: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:C::20: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:C::20: bytes=32 time<1ms TTL=125

Ping statistics for 2001:DB8:C::20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Etape 7 : Redistribution de routes

Scénario : Router 3 a une route statique vers un réseau externe (2001:DB8:EXT::/64).
Objectif : Redistribuer cette route dans OSPFv3.

Router3

```
Router3# configure terminal
(Création de la route statique vers le réseau externe)
Router3(config)# ipv6 route 2001:DB8:E::/64 2001:DB8:C::100
(Entrée dans le processus OSPFv3)
Router3(config)# ipv6 router ospf 1
(Redistribution de la route statique dans le domaine OSPF)
Router3(config-rtr)# redistribute static
Router3(config-rtr)# end
```

Router1

```
(Commande pour voir uniquement les routes apprises via OSPF)
Router1# show ipv6 route ospf
```

La table de routage sur R1 montre que tout fonctionne exactement comme prévu.

```
R1#show ipv6 route ospf
IPv6 Routing Table - 9 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
O   2001:DB8:B::/64 [110/2]
     via FE80::250:FFF:FE51:E102, GigabitEthernet0/1
O   2001:DB8:C::/64 [110/3]
     via FE80::250:FFF:FE51:E102, GigabitEthernet0/1
OE2 2001:DB8:E::/64 [110/20], tag 1
     via FE80::250:FFF:FE51:E102, GigabitEthernet0/1
O   2001:DB8:23::/64 [110/2]
     via FE80::250:FFF:FE51:E102, GigabitEthernet0/1
R1#
```